College of Science and Technology

Royal University of Bhutan



Bachelor of Engineering in Information Technology

CTE412 – Integrated System

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02230128

Task 1

1. Create a kind cluster named cte412-<sid> with one control-plane node and two worker nodes. Host port 30090, bound to 127.0.0.1 only, must reach node port 30090 on the cluster.

Kind-cluster.yaml file

```yaml
kind: Cluster
apiVersion: kind.x-k8s.io/v1alpha4
name: cte412-02230128
nodes:
  - role: control-plane
  - role: worker
  - role: worker
    extraPortMappings:
      - containerPort: 30090
        hostPort: 30090
        listenAddress: "127.0.0.1"
        protocol: TCP
```

Commands:

- kind create cluster --config cluster/kind-cluster.yaml

```
C:\Users\ASUS\Lab_14>kind create cluster --config cluster/kind-cluster.yaml
Creating cluster "cte412-02230128" ...
 • Ensuring node image (kindest/node:v1.37.0) 🖼 ...
 ✓ Ensuring node image (kindest/node:v1.37.0) 🖼
 • Preparing nodes 📦 📦 📦  ...
 ✓ Preparing nodes 📦 📦 📦
 • Writing configuration 📜 ...
 ✓ Writing configuration 📜
 • Starting control-plane 🕹 ...
 ✓ Starting control-plane 🕹
 • Installing CNI 🔌 ...
 ✓ Installing CNI 🔌
 • Installing StorageClass 💾 ...
 ✓ Installing StorageClass 💾
 • Joining worker nodes 🚜 ...
 ✓ Joining worker nodes 🚜
Set kubectl context to "kind-cte412-02230128"
You can now use your cluster with:

kubectl cluster-info --context kind-cte412-02230128

Thanks for using kind! 😊
```

2. Create the namespace nb-02230128 with the labels app.kubernetes.io/part-of=campusnoticeboard and cte412/practical="06", and an annotation recording your full name.

```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: nb-02230128
  labels:
    app.kubernetes.io/part-of: campus-notice-board
    cte412/practical: "06"
  annotations:
    owner: "Karma Sonam"
```

3. In that namespace, create a ResourceQuota with these hard limits: requests.cpu 1, requests.memory 1Gi, limits.cpu 2, limits.memory 2Gi, pods 15, count/services 4.

```yaml
apiVersion: v1
kind: ResourceQuota
metadata:
  name: nb-quota
  namespace: nb-02230128
spec:
  hard:
    requests.cpu: "1"
    requests.memory: "1Gi"
    limits.cpu: "2"
    limits.memory: "2Gi"
    pods: "15"
    count/services: "4"
```

4. Create a LimitRange supplying default requests and limits for containers. Choose the values yourself and justify each one in a comment against the quota above.

```yaml
apiVersion: v1
kind: LimitRange
metadata:
  name: nb-limits
  namespace: nb-02230128
spec:
  limits:
    - default:
        cpu: "200m"
        memory: "256Mi"
      defaultRequest:
        cpu: "100m"
        memory: "128Mi"
      type: Container
```

5. Make nb-<sid> the default namespace of your kubectl context.

```
C:\Users\ASUS\Lab_14>kubectl apply -f manifests/base/base.yaml
namespace/nb-02230128 created
resourcequota/nb-quota created
limitrange/nb-limits created

C:\Users\ASUS\Lab_14>kubectl config set-context --current --namespace=nb-02230128
Context "kind-cte412-02230128" modified.
```

Evidence:

```
NAME                              STATUS   ROLES           AGE   VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE                         KERNEL-VERSION
cte412-02230128-control-plane     Ready    control-plane   16m   v1.37.0   172.21.0.6    <none>        Debian GNU/Linux 13 (trixie)   6.6.87.2-microsoft-standard-WSL2
cte412-02230128-worker            Ready    <none>          15m   v1.37.0   172.21.0.5    <none>        Debian GNU/Linux 13 (trixie)   6.6.87.2-microsoft-standard-WSL2
cte412-02230128-worker2           Ready    <none>          15m   v1.37.0   172.21.0.7    <none>        Debian GNU/Linux 13 (trixie)   6.6.87.2-microsoft-standard-WSL2
```

```
Name:            nb-quota
Namespace:       nb-02230128
Resource         Used  Hard
--------         ----  ----
count/services   0     4
limits.cpu       0     2
limits.memory    0     2Gi
pods             0     15
requests.cpu     0     1
requests.memory  0     1Gi
```

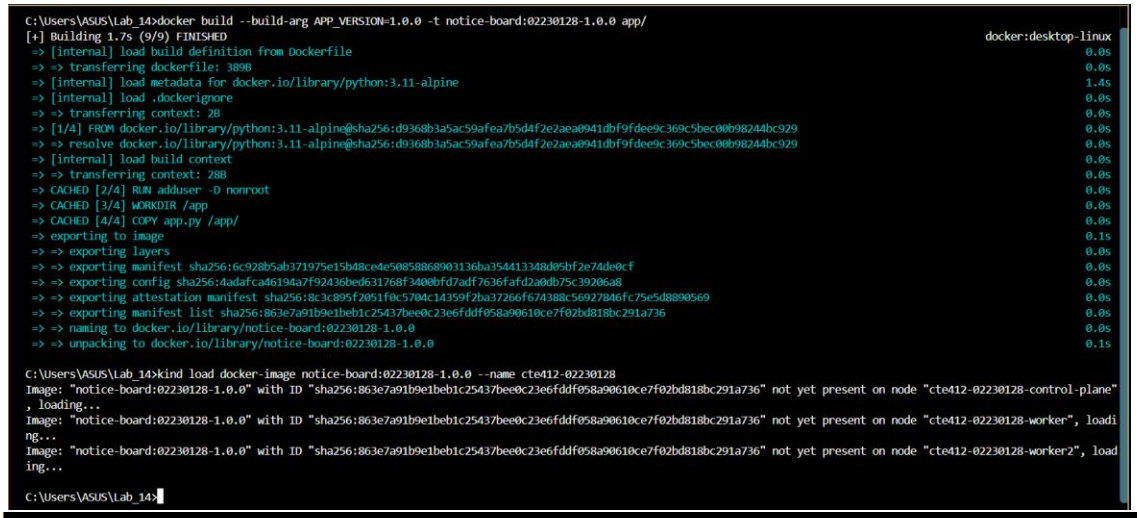
```
apiVersion: v1
clusters:
- cluster:
    certificate-authority-data: DATA+OMITTED
    server: https://127.0.0.1:60772
  name: kind-cte412-02230128
contexts:
- context:
    cluster: kind-cte412-02230128
    namespace: nb-02230128
    user: kind-cte412-02230128
  name: kind-cte412-02230128
current-context: kind-cte412-02230128
kind: Config
users:
- name: kind-cte412-02230128
  user:
    client-certificate-data: DATA+OMITTED
    client-key-data: DATA+OMITTED
```

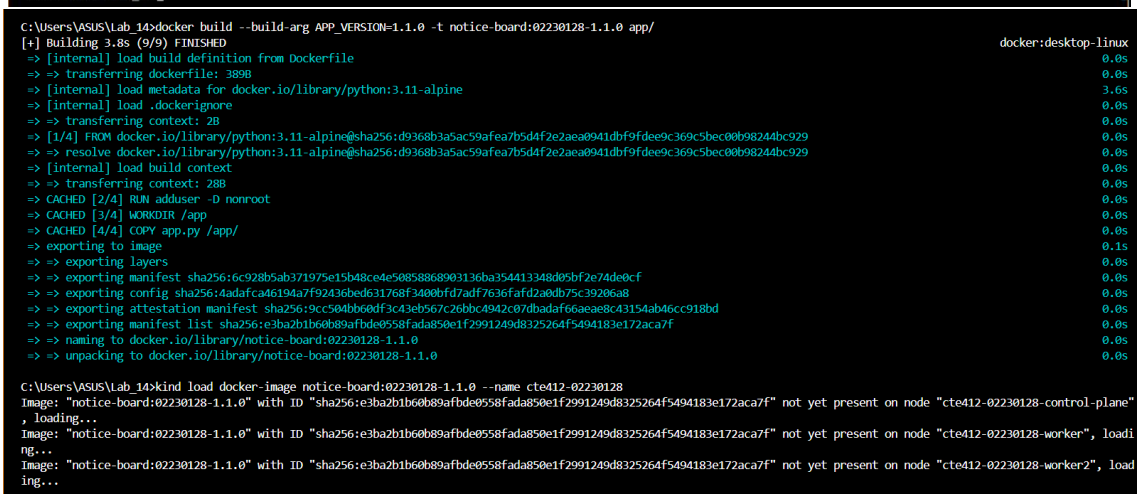
Task 2

1. Every version listen on port 8080, runs as a non-root user, and exposes three endpoints. GET /notices returns at least three notices as JSON. GET /version returns JSON containing version, student (your <sid>) and pod (the container hostname). GET /healthz returns HTTP 200 when healthy.
2. docker tag noticeboard-app:v1.0.0 notice-board:02230128-1.0.0
3. Tag notice-board:<sid>-1.1.0 — adds a category field to every notice and a pinned Boolean to at least one. When the environment variable FEATURE_PINNED equals

true, pinned notices are listed first and the response body includes "pinnedFirst": true; otherwise, the order is unchanged and the field is false.

4. Tag notice-board:<sid>-1.1.0 — adds a category field to every notice and a pinned Boolean to at least one. When the environment variable FEATURE_PINNED equals true, pinned notices are listed first and the response body includes "pinnedFirst": true; otherwise the order is unchanged and the field is false.
5. Tag notice-board:<sid>-1.2.0-broken — identical to 1.1.0 except that /healthz never returns 200. /notices and /version still work.
6. Place all three images into the cluster without using a registry.





```
C:\Users\ASUS\Lab_14>docker build --build-arg APP_VERSION=1.2.0-broken -t notice-board:02230128-1.2.0-broken app/
[+] Building 6.4s (9/9) FINISHED                                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                              0.0s
 => => transferring dockerfile: 389B                                                                                              0.0s
 => [internal] load metadata for docker.io/library/python:3.11-alpine                                                             6.2s
 => [internal] load .dockerignore                                                                                                 0.0s
 => => transferring context: 2B                                                                                                   0.0s
 => [1/4] FROM docker.io/library/python:3.11-alpine@sha256:d9368b3a5ac59afea7b5d4f2e2aea0941dbf9fdee9c369c5bec00b98244bc929        0.0s
 => => resolve docker.io/library/python:3.11-alpine@sha256:d9368b3a5ac59afea7b5d4f2e2aea0941dbf9fdee9c369c5bec00b98244bc929        0.0s
 => [internal] load build context                                                                                                 0.0s
 => => transferring context: 28B                                                                                                  0.0s
 => CACHED [2/4] RUN adduser -D nonroot                                                                                           0.0s
 => CACHED [3/4] WORKDIR /app                                                                                                     0.0s
 => CACHED [4/4] COPY app.py /app/                                                                                                0.0s
 => exporting to image                                                                                                            0.1s
 => => exporting layers                                                                                                           0.0s
 => => exporting manifest sha256:6c928b5ab371975e15b48ce4e5085886903136ba354413348d05bf2e74de0cf                                  0.0s
 => => exporting config sha256:4adafca46194a7f92436bed631768f3400bfd7adf7636fafd2a0db75c39206a8                                   0.0s
 => => exporting attestation manifest sha256:e8d8afd5431d853b9650a72736c5189e4366e61f7e693a6893e23b17ca4c5ff5                     0.0s
 => => exporting manifest list sha256:86a06b38fe8fec6dc09ea45116737d1ca568bc05dbd0603c88fac1b97e38c4f1                            0.0s
 => => naming to docker.io/library/notice-board:02230128-1.2.0-broken                                                             0.0s
 => => unpacking to docker.io/library/notice-board:02230128-1.2.0-broken                                                          0.0s

C:\Users\ASUS\Lab_14>kind load docker-image notice-board:02230128-1.2.0-broken --name cte412-02230128
Image: "notice-board:02230128-1.2.0-broken" with ID "sha256:86a06b38fe8fec6dc09ea45116737d1ca568bc05dbd0603c88fac1b97e38c4f1" not yet present on node "cte412-02230128-control
-plane", loading...
Image: "notice-board:02230128-1.2.0-broken" with ID "sha256:86a06b38fe8fec6dc09ea45116737d1ca568bc05dbd0603c88fac1b97e38c4f1" not yet present on node "cte412-02230128-worker"
, loading...
Image: "notice-board:02230128-1.2.0-broken" with ID "sha256:86a06b38fe8fec6dc09ea45116737d1ca568bc05dbd0603c88fac1b97e38c4f1" not yet present on node "cte412-02230128-worker2
", loading...
```

Evidence:

```
1   notice-board:02230128-1.0.0          863e7a91b9e1   92.9MB   23MB
2   notice-board:02230128-1.1.0          e3ba2b1b60b8   92.9MB   23MB
3   notice-board:02230128-1.2.0-broken   86a06b38fe8f   92.9MB   23MB
4
```

```
1   docker.io/library/notice-board   02230128-1.0.0          4adafca46194a   23MB
2   docker.io/library/notice-board   02230128-1.1.0          4adafca46194a   23MB
3   docker.io/library/notice-board   02230128-1.2.0-broken   4adafca46194a   23MB
```

Task 3 — Deploy and Expose Version 1.0.0 (25 minutes)

1. Create a Deployment notice-board running 3 replicas of 1.0.0, with labels app=noticeboard and version=<image version> on the Pods.
2. Declare requests and limits for the container that fit inside your quota with room for Tasks 5 to 8.
3. Configure a readiness probe against /healthz. Configure a liveness probe that does not use /healthz; state in a comment why.
4. Ensure the kubelet can never attempt a registry pull for this image.
5. Create a ClusterIP Service notice-board (port 80 to the named container port) and a NodePort. Service notice-board-ext on node port 30090.
6. From the host, show that /version on port 30090 is answered by at least two different Pods.

```
C:\Users\ASUS\Lab_14>kubectl apply -f manifests/rolling/task3-deploy.yaml
deployment.apps/notice-board created
service/notice-board created
service/notice-board-ext created
```

Evidence:

```
NAME                          READY   UP-TO-DATE   AVAILABLE   AGE     CONTAINERS     IMAGES                          SELECTOR
deployment.apps/notice-board  3/3     3            3           2m23s   notice-board   notice-board:02230128-1.0.0     app=notice-board

NAME                               READY   STATUS    RESTARTS   AGE     IP           NODE                      NOMINATED NODE   READINESS GATES
pod/notice-board-7c96dbdd6f-cpl74  1/1     Running   0          2m23s   10.244.2.2   cte412-02230128-worker2   <none>           <none>
pod/notice-board-7c96dbdd6f-pxfkp  1/1     Running   0          2m23s   10.244.1.2   cte412-02230128-worker    <none>           <none>
pod/notice-board-7c96dbdd6f-wmfnr  1/1     Running   0          2m23s   10.244.2.3   cte412-02230128-worker2   <none>           <none>
```

```
NAME                        TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
service/notice-board        ClusterIP   10.96.125.230   <none>        80/TCP         6m50s
service/notice-board-ext    NodePort    10.96.128.139   <none>        80:30090/TCP   6m50s

NAME                          ENDPOINTS                                                AGE
endpoints/notice-board        10.244.1.2:8080,10.244.2.2:8080,10.244.2.3:8080          6m50s
endpoints/notice-board-ext    10.244.1.2:8080,10.244.2.2:8080,10.244.2.3:8080          6m50s
```

```
StatusCode        : 200
StatusDescription : OK
Content           : {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cpl74"}
RawContent        : HTTP/1.0 200 OK
                    Content-Type: application/json
                    Date: Mon, 19 Oct 2026 13:48:39 GMT
                    Server: BaseHTTP/0.6 Python/3.11.17

                    {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f...
Forms             : {}
Headers           : {[Content-Type, application/json], [Date, Mon, 19 Oct 2026 13:48:39 GMT], [Server, BaseHTTP/0.6
                    Python/3.11.17]}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 83
```

Task 4 — Availability Monitor and Summariser (15 minutes)

The monitor is the measuring instrument for Tasks 5 to 8. It must run continuously and must not be restarted between measurements.

1. Create a Pod monitor in manifests/base/ that requests http://notice-board/version every 0.5 seconds, forever, and writes one line per request to standard output in the form <ISO-8601 timestamp> <OK|FAIL> <response body or error>.
2. Write scripts/summarise.sh. Given a saved monitor log as its only argument, it prints:
   total requests, failed requests, the first and last timestamp, and the count of responses per distinct version value.
3. Save one minute of monitor output with the system at rest and summarize it.

```
C:\Users\ASUS\Lab_14>kubectl apply -f manifests/base/monitor-pod.yaml
pod/availability-monitor created

C:\Users\ASUS\Lab_14>
```

Evidence:

```
2026-10-19T14:06:20.625323Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
2026-10-19T14:06:21.424732Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cpl74"}
2026-10-19T14:06:22.022949Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
2026-10-19T14:06:22.707616Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
2026-10-19T14:06:23.232868Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cpl74"}
2026-10-19T14:06:23.811885Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
2026-10-19T14:06:24.418072Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cpl74"}
2026-10-19T14:06:25.012158Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
2026-10-19T14:06:25.709457Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-pxfkp"}
2026-10-19T14:06:26.238449Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
2026-10-19T14:06:26.918826Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-pxfkp"}
2026-10-19T14:06:27.512961Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
2026-10-19T14:06:28.215965Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
2026-10-19T14:06:28.744755Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
```

```
=== Availability Log Summary ===
Log File:         evidence/e4.1-at-rest-log.txt
Start Timestamp:  2026-10-19T14:06:20.625323Z
End Timestamp:    2026-10-19T14:07:29.923491Z
Total Requests:   114
Successful (OK):  114
Failed (FAIL):    0

=== Version Counts ===
    114 "version": "1.0.0"
================================
```

Task 5

1. Set the Deployment strategy to Recreate. Release 1.1.0 by editing and applying the manifest. Record a change-cause for the revision.
2. Return to 1.0.0 declaratively.
3. Set the strategy to RollingUpdate with at most zero unavailable Pods, at most one surge Pod, and a minimum ready time of 5 seconds. Release 1.1.0 again. Record a changecause.
4. From each log, identify the mixed-version window: the interval during which both 1.0.0 and 1.1.0 responses were observed.

```
ASUS@Karma MINGW64 ~/Lab_14
$ kubectl logs availability-monitor -f > evidence/e5.1-recreate-log.txt &
MON_PID=$!
[1] 1571
```

```
ASUS@Karma MINGW64 ~/Lab_14
$ kubectl apply -f manifests/rolling/task5-recreate.yaml
sleep 25
deployment.apps/notice-board configured

ASUS@Karma MINGW64 ~/Lab_14
$ kubectl set image deployment/notice-board notice-board=notice-board:02230128-1.0.0
sleep 25
kill $MON_PID
deployment.apps/notice-board image updated
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]

ASUS@Karma MINGW64 ~/Lab_14
$
```

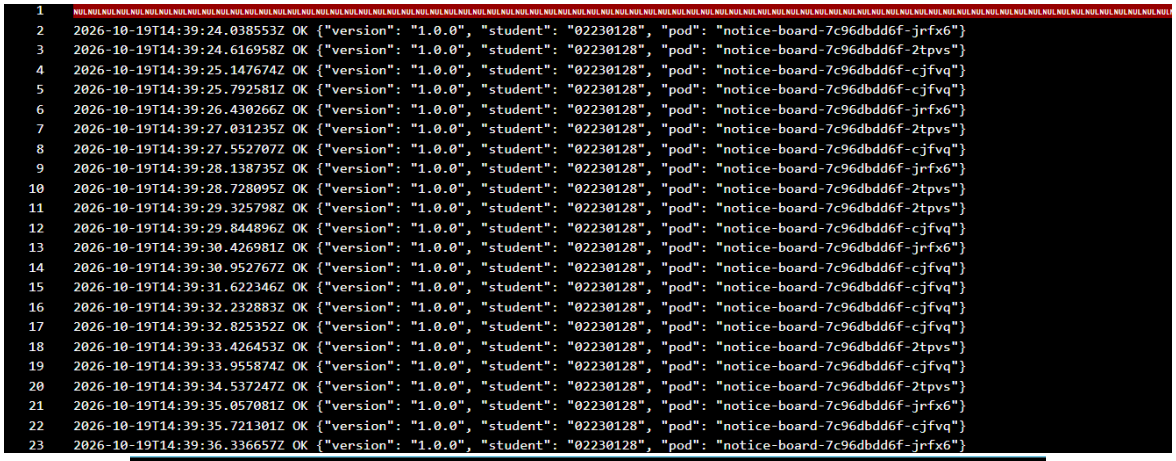
```
ASUS@Karma MINGW64 ~/Lab_14
$ ./scripts/summarise.sh evidence/e5.1-recreate-log.txt > evidence/e5.1-recreate-summary.txt
```

Evidence:

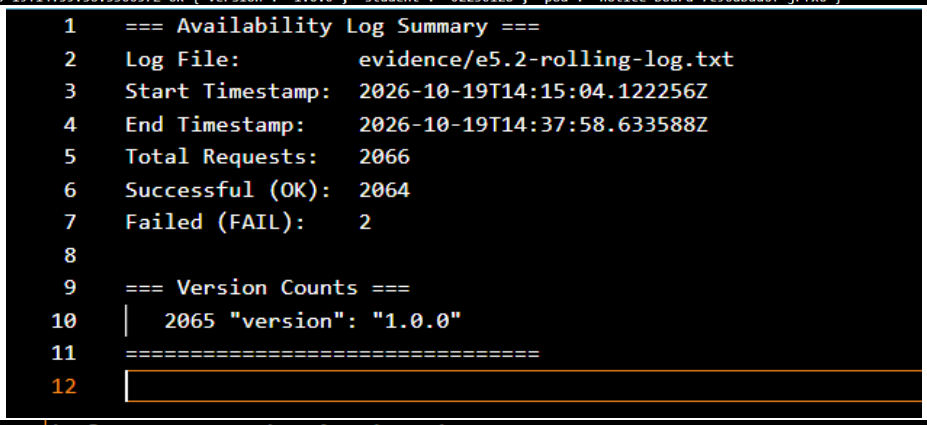
```
2026-10-19T14:15:04.122256Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cpl74"}
2026-10-19T14:15:04.740600Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cpl74"}
2026-10-19T14:15:05.523411Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
2026-10-19T14:15:06.225534Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
2026-10-19T14:15:06.822770Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cpl74"}
2026-10-19T14:15:07.620685Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
2026-10-19T14:15:08.323156Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cpl74"}
2026-10-19T14:15:08.950408Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cpl74"}
2026-10-19T14:15:09.558086Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cpl74"}
2026-10-19T14:15:10.240207Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cpl74"}
2026-10-19T14:15:11.021730Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
2026-10-19T14:15:11.551025Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-pxfkp"}
2026-10-19T14:15:12.151927Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cpl74"}
2026-10-19T14:15:12.825068Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-pxfkp"}
2026-10-19T14:15:13.356948Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-pxfkp"}
2026-10-19T14:15:14.041176Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cpl74"}
2026-10-19T14:15:14.730520Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-wmfnr"}
```

```
=== Availability Log Summary ===
Log File:         evidence/e5.1-recreate-log.txt
Start Timestamp:  2026-10-19T14:15:04.122256Z
End Timestamp:    2026-10-19T14:26:37.968965Z
Total Requests:   914
Successful (OK):  912
Failed (FAIL):    2

=== Version Counts ===
   913 "version": "1.0.0"
================================
```

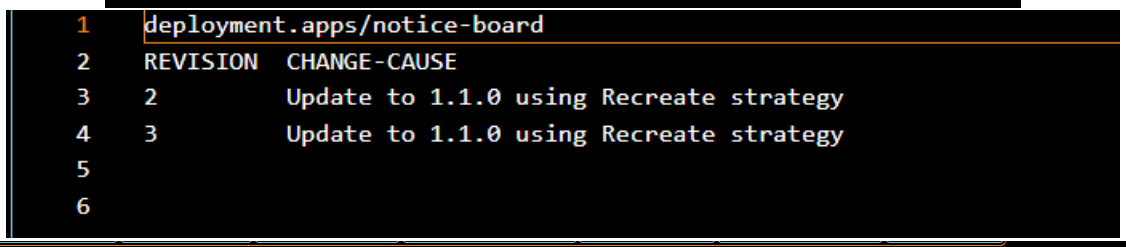


```
2026-10-19T14:39:24.038553Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-jrfx6"}
2026-10-19T14:39:24.616958Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-2tpvs"}
2026-10-19T14:39:25.147674Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cjfvq"}
2026-10-19T14:39:25.792581Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cjfvq"}
2026-10-19T14:39:26.430266Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-jrfx6"}
2026-10-19T14:39:27.031235Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-2tpvs"}
2026-10-19T14:39:27.552707Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cjfvq"}
2026-10-19T14:39:28.138735Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-jrfx6"}
2026-10-19T14:39:28.728095Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-2tpvs"}
2026-10-19T14:39:29.325798Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-2tpvs"}
2026-10-19T14:39:29.844896Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cjfvq"}
2026-10-19T14:39:30.426981Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-jrfx6"}
2026-10-19T14:39:30.952767Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cjfvq"}
2026-10-19T14:39:31.622346Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cjfvq"}
2026-10-19T14:39:32.232883Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cjfvq"}
2026-10-19T14:39:32.825352Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cjfvq"}
2026-10-19T14:39:33.426453Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-2tpvs"}
2026-10-19T14:39:33.955874Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cjfvq"}
2026-10-19T14:39:34.537247Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-2tpvs"}
2026-10-19T14:39:35.057081Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-jrfx6"}
2026-10-19T14:39:35.721301Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-cjfvq"}
2026-10-19T14:39:36.336657Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-7c96dbdd6f-jrfx6"}
```

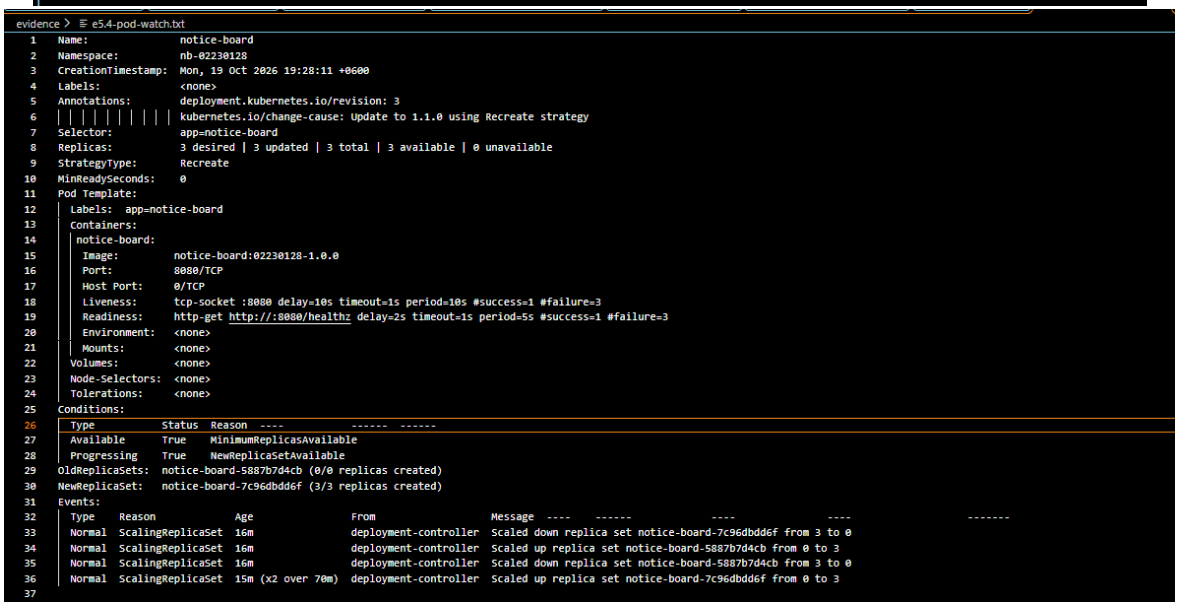


```
=== Availability Log Summary ===
Log File:          evidence/e5.2-rolling-log.txt
Start Timestamp:   2026-10-19T14:15:04.122256Z
End Timestamp:     2026-10-19T14:37:58.633588Z
Total Requests:    2066
Successful (OK):   2064
Failed (FAIL):     2

=== Version Counts ===
  2065 "version": "1.0.0"
==================================
```



```
deployment.apps/notice-board
REVISION  CHANGE-CAUSE
2         Update to 1.1.0 using Recreate strategy
3         Update to 1.1.0 using Recreate strategy
```



evidence > e5.4-pod-watch.txt

```
Name:               notice-board
Namespace:          nb-02230128
CreationTimestamp:  Mon, 19 Oct 2026 19:28:11 +0600
Labels:             <none>
Annotations:        deployment.kubernetes.io/revision: 3
                    kubernetes.io/change-cause: Update to 1.1.0 using Recreate strategy
Selector:           app=notice-board
Replicas:           3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:       Recreate
MinReadySeconds:    0
Pod Template:
  Labels:  app=notice-board
  Containers:
   notice-board:
    Image:         notice-board:02230128-1.0.0
    Port:          8080/TCP
    Host Port:     0/TCP
    Liveness:      tcp-socket :8080 delay=10s timeout=1s period=10s #success=1 #failure=3
    Readiness:     http-get http://:8080/healthz delay=2s timeout=1s period=5s #success=1 #failure=3
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  notice-board-5887b7d4cb (0/0 replicas created)
NewReplicaSet:   notice-board-7c96dbdd6f (3/3 replicas created)
Events:
  Type    Reason             Age                From                   Message
  ----    ------             ----               ----                   -------
  Normal  ScalingReplicaSet  16m                deployment-controller  Scaled down replica set notice-board-7c96dbdd6f from 3 to 0
  Normal  ScalingReplicaSet  16m                deployment-controller  Scaled up replica set notice-board-5887b7d4cb from 0 to 3
  Normal  ScalingReplicaSet  16m                deployment-controller  Scaled down replica set notice-board-5887b7d4cb from 3 to 0
  Normal  ScalingReplicaSet  15m (x2 over 70m)  deployment-controller  Scaled up replica set notice-board-7c96dbdd6f from 0 to 3
```

Task 6

1. Set the Deployment so that a rollout making no progress for 60 seconds is marked failed. Release 1.2.0-broken declaratively, with a change-cause.
2. Show that Kubernetes itself reported the rollout as failed, and state the condition type and reason it reported.
3. Roll back to the previous revision. This is the only imperative state change permitted in the practical.
4. Bring the repository manifest back into agreement with the live Deployment, and prove that they agree.
5. The monitor must record zero failures between the broken apply and the completed rollback. If it records any, your configuration is wrong: correct it and repeat the task.

```
ASUS@Karma MINGW64 ~/Lab_14
$ kubectl apply -f manifests/rolling/task6-broken.yaml
deployment.apps/notice-board configured
```

```
ASUS@Karma MINGW64 ~/Lab_14
$ kubectl rollout status deployment/notice-board
deployment "notice-board" successfully rolled out
```

Evidence:

```
2026-10-19T14:59:27.111933Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-9gpsx"}
2026-10-19T14:59:27.792685Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-9gpsx"}
2026-10-19T14:59:28.390592Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-bxljh"}
2026-10-19T14:59:28.918629Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-bxljh"}
2026-10-19T14:59:29.611357Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-cvrkp"}
2026-10-19T14:59:30.132343Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-cvrkp"}
2026-10-19T14:59:30.665819Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-cvrkp"}
2026-10-19T14:59:31.232980Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-9gpsx"}
2026-10-19T14:59:31.824242Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-bxljh"}
2026-10-19T14:59:32.421327Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-bxljh"}
2026-10-19T14:59:33.105633Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-cvrkp"}
2026-10-19T14:59:33.624165Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-cvrkp"}
2026-10-19T14:59:34.145332Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-9gpsx"}
2026-10-19T14:59:34.726009Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-bxljh"}
2026-10-19T14:59:35.245179Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-bxljh"}
2026-10-19T14:59:35.766482Z OK {"version": "1.0.0", "student": "02230128", "pod": "notice-board-57d978b75-cvrkp"}
```

```
Available   True    MinimumReplicasAvailable    Deployment has minimum availability.
Progressing True    NewReplicaSetAvailable  ReplicaSet "notice-board-7c96dbdd6f" has successfully progressed.

```

```
NAME                            READY   STATUS    RESTARTS   AGE     IP            NODE                        NOMINATED NODE   READINESS GATES
availability-monitor            1/1     Running   0          8m16s   10.244.1.7    cte412-02230128-worker      <none>           <none>
notice-board-7c96dbdd6f-2v524   1/1     Running   0          95s     10.244.1.9    cte412-02230128-worker      <none>           <none>
notice-board-7c96dbdd6f-ch57t   1/1     Running   0          79s     10.244.2.11   cte412-02230128-worker2     <none>           <none>
notice-board-7c96dbdd6f-j6kl2   1/1     Running   0          112s    10.244.2.10   cte412-02230128-worker2     <none>           <none>

```

```
=== Availability Log Summary ===
Log File:         evidence/t6-monitor.log
Start Timestamp:  2026-10-19T14:59:27.111933Z
End Timestamp:    2026-10-19T15:05:17.016220Z
Total Requests:   603
Successful (OK):  603
Failed (FAIL):    0

=== Version Counts ===
   603 "version": "1.0.0"
================================

```

Task 7

1. In manifests/bluegreen/, create Deployments notice-board-blue (1.0.0) and noticeboardgreen (1.1.0), each with 3 replicas, the same probes and resources as Task 3, and a Pod label slot with value blue or green.
2. The production Services (notice-board and notice-board-ext) send traffic to blue only. A third Service, notice-board-preview, always sends traffic to green.
3. Verify green through the preview Service from inside the cluster while production still serves blue.
4. Switch production to green by editing and applying the Service manifest(s). Then switch back to blue, then to green again.
5. Capture the monitor log across all three switches.

```
ASUS@Karma MINGW64 ~/Lab_14
$ kubectl apply -f manifests/bluegreen/
deployment.apps/notice-board-blue created
deployment.apps/notice-board-green created
service/notice-board created
service/notice-board-preview created
```

```
ASUS@Karma MINGW64 ~/Lab_14
$ kubectl apply -f manifests/base/monitor-pod.yaml
kubectl logs availability-monitor -f > evidence/e7.3-switch-log.txt &
MON_PID=$!
sleep 15
pod/availability-monitor created
[1] 2089
container "monitor" in pod "availability-monitor" is waiting to start: ContainerCreating
container "monitor" in pod "availability-monitor" is waiting to start: ContainerCreating
container "monitor" in pod "availability-monitor" is waiting to start: ContainerCreating
container "monitor" in pod "availability-monitor" is waiting to start: ContainerCreating
container "monitor" in pod "availability-monitor" is waiting to start: ContainerCreating
container "monitor" in pod "availability-monitor" is waiting to start: ContainerCreating
container "monitor" in pod "availability-monitor" is waiting to start: ContainerCreating
container "monitor" in pod "availability-monitor" is waiting to start: ContainerCreating
container "monitor" in pod "availability-monitor" is waiting to start: ContainerCreating
container "monitor" in pod "availability-monitor" is waiting to start: ContainerCreating
container "monitor" in pod "availability-monitor" is waiting to start: ContainerCreating
```

Evidence:

```
1   NAME            ENDPOINTS                                AGE
2   notice-board    10.244.1.11:8080,10.244.2.12:8080    4m40s
3
```

evidence > e7.3-switch-summary.txt

```
1   === Availability Log Summary ===
2   Log File:         evidence/e7.3-switch-log.txt
3   Start Timestamp:  /bin/sh:
4   End Timestamp:    2026-10-19T15:22:08.608163Z
5   Total Requests:   478
6   Successful (OK):  0
7   Failed (FAIL):    239
8
9   === Version Counts ===
10  ================================
11
```

```
1   Name:             nb-quota
2   Namespace:        nb-02230128
3   Resource          Used    Hard
4   --------          ----    ----
5   count/services    2       4
6   limits.cpu        1200m   2
7   limits.memory     1536Mi  2Gi
8   pods              6       15
9   requests.cpu      600m    1
10  requests.memory   768Mi   1Gi
11
```

Task 8

1. Create a ConfigMap notice-board-flags holding FEATURE_PINNED with the value false, and make the green Deployment read the variable from it.
2. Change the flag to true in the manifest and apply it. Determine, with evidence, whether the running green Pods observe the change.
3. Make the new flag value take effect on all green Pods with zero failed requests, using only declarative changes.

```
ASUS@Karma MINGW64 ~/Lab_14
$ kubectl apply -f manifests/bluegreen/configmap.yaml
configmap/notice-board-flags created
```

```
ASUS@Karma MINGW64 ~/Lab_14
$
$ kubectl apply -f manifests/bluegreen/green-deployment.yaml
deployment.apps/notice-board-green configured
```

```
ASUS@Karma MINGW64 ~/Lab_14
$ kubectl apply -f manifests/bluegreen/configmap.yaml
kubectl rollout restart deployment/notice-board-green -n nb-02230128
kubectl rollout status deployment/notice-board-green -n nb-02230128
configmap/notice-board-flags configured
deployment.apps/notice-board-green restarted
Waiting for deployment "notice-board-green" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "notice-board-green" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "notice-board-green" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "notice-board-green" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "notice-board-green" rollout to finish: 1 old replicas are pending termination...
deployment "notice-board-green" successfully rolled out
```

Evidence:


```
[{"id": 1, "title": "Welcome to CST", "content": "Lab session active."}, {"id": 2, "title": "Cluster Maintenance", "content": "Scheduled tonight."},
```



```
[{"id": 1, "title": "Welcome to CST", "content": "Lab session active."}, {"id": 2, "title": "Cluster Maintenance", "content": "Scheduled tonight."},
```


```
1    === Availability Log Summary ===
2    Log File:         evidence/e7.3-switch-log.txt
3    Start Timestamp:  /bin/sh:
4    End Timestamp:    2026-10-19T15:22:08.608163Z
5    Total Requests:   478
6    Successful (OK):  0
7    Failed (FAIL):    239
8
9    === Version Counts ===
10   ================================
```

Task 9

1. Save the complete monitor log for the session as evidence/t9-monitor-full.log.
2. Complete the README rebuild sequence, then prove it: delete every object in nb-<sid> except the namespace, quota and limit range, and recreate the final state of Task 8 using the README commands only.
3. Reset the context namespace to default, delete the cluster, and commit.

```
ASUS@Karma MINGW64 ~/Lab_14
$ kubectl apply -f manifests/base/
kubectl apply -f manifests/bluegreen/
kubectl get all -n nb-02230128 > evidence/e9.1-rebuild-output.txt
namespace/nb-02230128 unchanged
resourcequota/nb-quota unchanged
limitrange/nb-limits unchanged
pod/availability-monitor unchanged
deployment.apps/notice-board-blue unchanged
configmap/notice-board-flags unchanged
deployment.apps/notice-board-green unchanged
service/notice-board configured
service/notice-board-preview unchanged
```

```
[+] Building 3.6s (8/8) FINISHED                                                
```